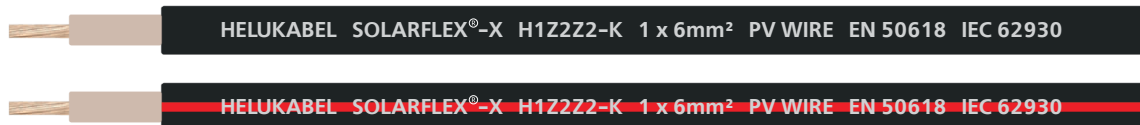


SOLARFLEX®-X H1Z2Z2-K

1500 V DC



TECHNICAL DATA

Cross-linked single core cable acc. to DIN VDE 0283-618 / DIN EN 50618

Temperature range	fixed -40°C to +90°C
Permissible operating temperature of the conductor	+120°C
Nominal voltage	AC U ₀ /U 1000/1000 V DC U ₀ /U 1500/1500 V
Test voltage	6500 V
Minimum bending radius	fixed 5x Outer-Ø

CABLE STRUCTURE

- Copper wire tinned, finely stranded acc. to DIN VDE 0295 Class 5 / IEC 60228 Class 5
- Core insulation made of low smoke halogen free E-beam irradiation crosslinked polyolefin (XLPO) insulation
- Outer sheath made of low smoke halogen free E-beam irradiation crosslinked polyolefin (XLPO) sheath
- Sheath colour: see table below

PROPERTIES

- resistant to: UV radiation, ozone, weathering effects, water
- for outdoor use
- direct burial
- halogen-free

TESTS

- halogen-free acc. to DIN VDE 0482-754-1 / DIN EN 60754-1 / IEC 60754-1
- flame-retardant acc. to DIN VDE 0482-332-1-2 / DIN EN 60332-1-2 / IEC 60332-1-2
- smoke density acc. to DIN VDE 0482-1034-1+2 / DIN EN 61034-1+2 / IEC 61034-1+2
- UV-resistant acc. to DIN VDE 0283-618 / DIN EN 50618 Appendix E
- ozone-resistant acc. to DIN VDE 0283-618 / DIN EN 50618
- weather-resistant acc. to DIN VDE 0283-618 / DIN EN 50618 Appendix E
- water-resistant acc. to DIN VDE 0285-525-2-21 / DIN EN 50525-2-21 Appendix E
- DC Voltage resistance of the insulation acc. to DIN VDE 0283-618 / DIN EN 50618 Tab. 2
- Certifications: TÜV Rheinland IEC 62930, EN 50618

APPLICATION

SOLARFLEX®-X H1Z2Z2-K is used for wiring solar modules. Suitable for direct burial; recommendation: laying in pipes. Not suitable for permanent installation in water. The maximum permissible DC voltage of the system in which the cable is installed must not exceed 1.8 kV. The cable is suitable for use in and on devices and systems with protective insulation (protection class II).

NOTES

- the conductor is metrically (mm²) constructed, AWG numbers are approximated, and are for reference only

No. cores x cross-sec. mm ²	AWG, approx.	Outer-Ø min - max mm	Conductor Ø app. mm	Weight kg/km, approx.	black	black with red stripe
					Part no.	Part no.
1 x 2.5	14	5.1	1.95	45.0	18054265	-
1 x 4	12	5.5	2.45	60.0	18054160	18054294
1 x 6	10	6.1	3.00	80.0	18054161	18054292
1 x 10	8	7.2	4.00	126.0	18054266	-
1 x 16	6	8.8	5.35	170.0	18054267	-
1 x 25	4	10.8	6.80	270.0	18054268	-
1 x 35	2	12.4	8.10	365.0	18054269	-

No. cores x cross-sec. mm ²	Current carrying capacity of PV cables in accordance to the installation (T=60°C)			Max conductor DC resistance
	Single cable free in air (A)	Single cable on surface (A)	2 cable adjacent on surface (A)	Tinned copper (Ω/km)
1 x 2.5	42	40	33	8.21
1 x 4	57	54	45	5.09
1 x 6	72	69	58	3.39
1 x 10	98	96	80	1.95
1 x 16	132	130	107	1.24
1 x 25	183	174	138	0.795
1 x 35	227	215	171	0.565

APPENDIX

Use and type of installation for application in photovoltaic (HD 60364-7-712). For fixed installation indoors and outdoors. Installation of walls, walkways, pipes, conduits, and similar systems. The cables are suitable for use with Class II

CHEMICAL PROPERTIES

Halogen-free	acc. to EN 50525-1 Annex B (EN 50267-2-1, EN 50267-2-2, IEC 60754-1, IEC 60754-2)
Low Smoke Emission	acc. to IEC 61034, EN 61034 (Light Transmittance > 60%)
Weather resistance	Ozone resistance: acc. to EN 60811-403 Test Method A, EN 50396 clause 8.1.3 Test Method B Weathering/UV resistance: acc. to EN 50618 Annex E, EN 50289-4-17 (Method A), EN ISO 4892-1/-2 tensile strength and elongation at break after 720h (360 Cycles) of exposure to UV lights
Acid and alkaline resistance	acc. to EN 50618:2014 Annex B: 7 days, 23°C (N-Oxalic Acid, N-Sodium Hydroxide) as for EN 60811-404
Resistance to fire	Flame propagation acc. to EN 60332-1-2 (Single Cable Flame Test) According to: EN 50399 Common test methods for cables under fire conditions Heat release and smoke production measurement on cables during flame spread test, UNI EN 13051-6 Flammability class: Dca Smoke emission class: s2 Drip particle class: d2

MECHANICAL PROPERTIES

Tensile strength and elongation	for insulation and sheath before ageing acc. to EN 50618 Annex B (test acc. To EN 60811-501). tensile strength $\geq 8 \text{ N/mm}^2$ elongation at break for insulation and sheath $\geq 125\%$
Shrinkage test on sheath	acc. to EN 50618, Table 2: <2% (test acc. to EN 60811-503)
Dynamic Penetration Test	acc. to EN 50618
Durability of Print	acc. to EN 50618 (test acc. to EN 50396)
Direct Burial	Impact test resistance of single conductor type USE and USE-2 cables (acc. to UL854) Rodent resistance safety can be optimized by utilizing protective hoses and cables with spinning or braid metallic coatings
Long term resistance of insulation to d.c	acc. to EN 50618, Table 2 test acc. to EN 50395 clause 9: Cable immersed in water containing 1% NaCl for 240h ; Water temperature: $85^\circ\text{C} \pm 5$; Voltage applied: 1.8 kV D.C.
Water resistance	AD7 category acc. to DIN VDE 0285-525-2-21 / DIN EN 50525-2-21 Appendix E

THERMAL PROPERTIES

Lifetime	acc. to EN 50618 : 25 years the cables are designed to operate at a normal max conductor temperature of 90°C , but for a maximum of 20,000 hours with conductor temperature of 120°C at a max. ambient temperature of 90°C is permitted. (test according to EN 60216-1 and EN 60216-2)
Max. short circuit temperature	250°C (for 5 sec.)
Resistance to cold	EN 50618, Table 2: Cold Bending Test at -40°C acc. to EN 60811-504; Cold Elongation Test at -40°C acc. to EN 60811-505; Cold Impact Test at -40°C acc. to EN 50618 Annex C and EN 60811-506. Damp-Heat Test Acc. to EN 50618, Table 2 (test acc. to EN 60068-2-78) : 90°C for 1,000h and min. 85% humidity